

Multiple Choice (10 marks)

1. The practice range at a golf course charges \$4.00 for a bucket of 40 golf balls. At this rate, how much will a bucket of 100 golf balls cost?
 - (a) \$10.00
 - (b) \$10.50
 - (c) \$13.50
 - (d) \$16.00
2. Find the median in the set of data 23, 13, 18, 29, 32, 25.
 - (a) 18
 - (b) 24
 - (c) 25
 - (d) 29
3. Which expression is equivalent to $5(4x + 3) - 2x$?
 - (a) $18x + 15$
 - (b) $18x + 3$
 - (c) $7x + 8$
 - (d) $2x + 8$
4. What is the value of the expression $2(3(4^2 + 1)) - 2^3$?
 - (a) 156
 - (b) 110
 - (c) 94
 - (d) 48
5. Which group of numbers is in order from least to greatest?
 - (a) 0.25 1.6 1.0
 - (b) 1.0 0.25 1.6
 - (c) 0.25 1.0 1.6
 - (d) 1.6 1.0 0.25

True / False (10 marks)

Answer True or False

6. True or False: A correlation of 0.6 indicates that the percentage of variation in y explained by x is twice that of a correlation of 0.3.
7. True or False: Tara will take 1 hour and 12 minutes to develop 8 rolls of film at her current rate.
8. True or False: It takes approximately 12.68 years for two thirds of a radioactive substance with a half-life of 8 years to decay.
9. True or False: Marcus took 16 minutes to wash his car at the car wash using his quarters.
10. True or False: Joe finished the race at 3:58 p.m., which is 4 minutes before Mike's finish time of 4:02 p.m.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. We define two sequences of vectors $(\mathbf{v}_n)$ and $(\mathbf{w}_n)$ as follows: First, $\mathbf{v}_0 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, $\mathbf{w}_0 = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$. Then for all $n \geq 1$, $\mathbf{v}_n$ is the projection of $\mathbf{w}_{n-1}$ onto $\mathbf{v}_0$, and $\mathbf{w}_n$ is the projection of $\mathbf{v}_n$ onto $\mathbf{w}_0$. Find

$$\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 + \cdots .$$

12. Calculate the total cost of purchasing 4 magazines at 2.99 each and 3 books at 6.99 each. Explain your calculation process and the importance of budgeting in this scenario.

End of Paper